Counting the five never-counted index-calculus items

research/ic_acct7_complete_counters_20261107 · 2026-11-07 · branch claude/ic-complete-counters · class: accounting

Question

Item 6 (research/ic_acct6_archive_20261021) found 22,706 archive records pricing incomplete in bitop/v1 because the code never counted five index-calculus items: the elimination's scans and pivot inversions, relation row construction, table insert attempts, walk bookkeeping and base index builds. Its two re-runs of ic.shared_rank sessions stayed incomplete because that method wrote only one of the five markers. This change counts the items where the current code still does not, and checks which archive configurations now write every marker. It changes no measured cost.

What the code now counts

ic.shared_rank writes every marker, each counter at the site that does the work: the folded table's insert attempts (oracle_setup), the matrix's entry scans and pivot-inversion steps after the full-rank check's reads (linear_algebra), every 64-bit word its StdRng hands out through a forwarding wrapper that leaves the stream unchanged (relations, target_query), the rows' words, sums and h·a products (linear_algebra, target_relation_check), the target's curve-equation check, and the descent's column-log combination as two new counters, combination_modr_muls and combination_modr_adds (BITOPS.md "Extensions", 2026-11-07; no primitive weight changed; the frozen vectors and the work-v1 fixture tests pass). The gaudry-d4 and gaudry-d4-bin oracles build no table in their set-up, so they now say so with the table_insert_attempts marker; with large primes on, the binary oracle's pivot store is uncounted and the item stays listed.

Which archive configurations write every marker

The 106 distinct IC configurations in item 6's 396 session specs were run on toy curves of the same kind by an #[ignore] diagnostic test (a check of code paths, not a measurement). Figure 1 counts, over the 90 configurations both runs covered, those whose record still lists each item.

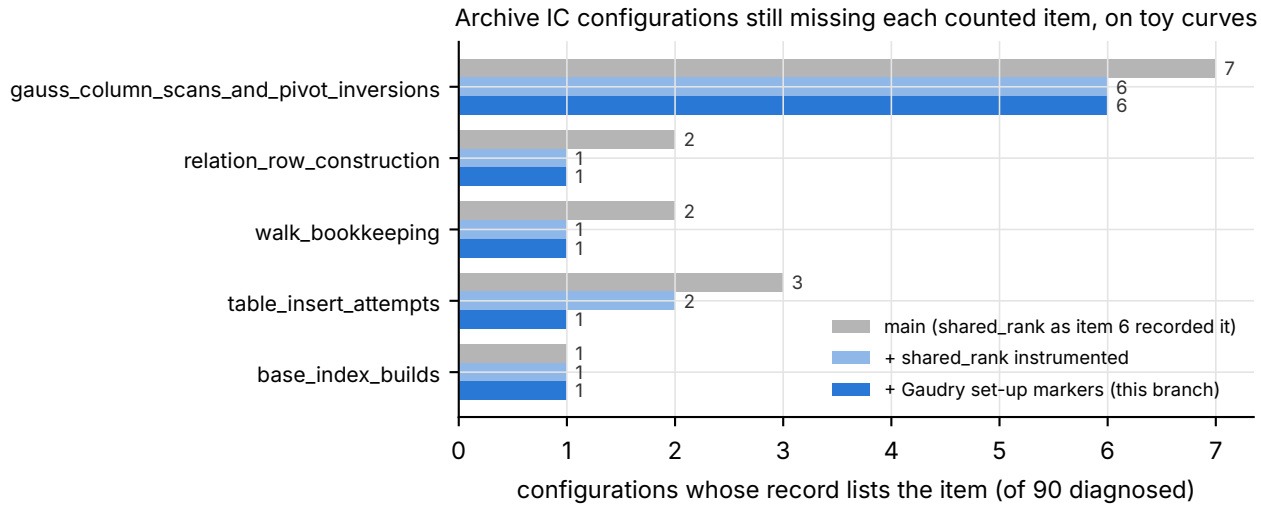


Figure 1: Configurations 1–90 of coverage/configs.jsonl whose record lists each item. "main" is the intermediate run with the ic.shared_rank row as item 6's tranche recorded it on main (four items). Source: coverage/items.tsv, coverage-intermediate.tsv, coverage-final.tsv.

Of the 97 configurations run on this branch, 68 price complete and 10 do not: five deferred sparse solvers (sparse-cado, wiedemann, lanczos, filter+*, whose SparseRows::solve has no per-site counters, so the elimination item stays), ic.k4_gaudry and ic.large_prime (their own pipelines count none of the five), gaudry-d4-bin (all five counted; solver gaps remain) and two cyclotomic bases (an unpriceable size counter). 19 could not run inside the library test (ecbench-binary plug-ins, absent ic.cm_folded_* native backends, one curve without an invariant subspace); nine slow gaudry-d4 configurations over F_{p^4} were not run.

The committed archive, before and after

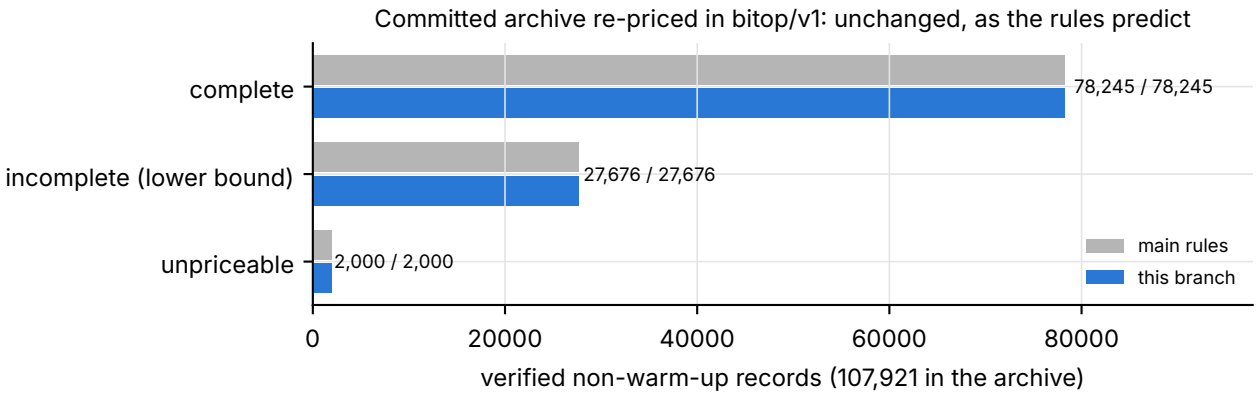


Figure 2: Every `research/**/records.jsonl` at main `a2be66091` (876 session directories, 123 unreadable, 107,921 verified non-warm-up records) priced by `examples/ic_acct6_archive.rs breakdown` with main's rules and this branch's. Source: `breakdown-before.json`, `breakdown-after.json`, `BEFORE_AFTER.md`.

	before	after
complete	78,245	78,245
incomplete	27,676	27,676
unpriceable	2,000	2,000

The breakdowns are identical apart from their label, as they must be: a committed record carries the markers its binary wrote, and the only pricing change is two rules for counter names no committed record carries. Records become complete only by re-running on this binary; no leaderboard row's pricing changed.

Re-run tranche

The 24 re-runnable sessions whose incomplete configurations all have a complete record somewhere (item 6′s PLAN.md, about 2.0 h) were not run: this host has no AWS identity and the resume-checkpoint gate refuses without an upload token, the stated stop conditions. No archive record is superseded.

Limits

Coverage was checked on toy curves, not on the sessions' curves. The sparse solvers, `ic.k4_gaudry`, `ic.large_prime` and `ic.cm_folded_*` keep their items, and `check_full_rank_base_columns`'s representative scan (one word a base point) is not counted in `ic.shared_rank`'s verification phase.